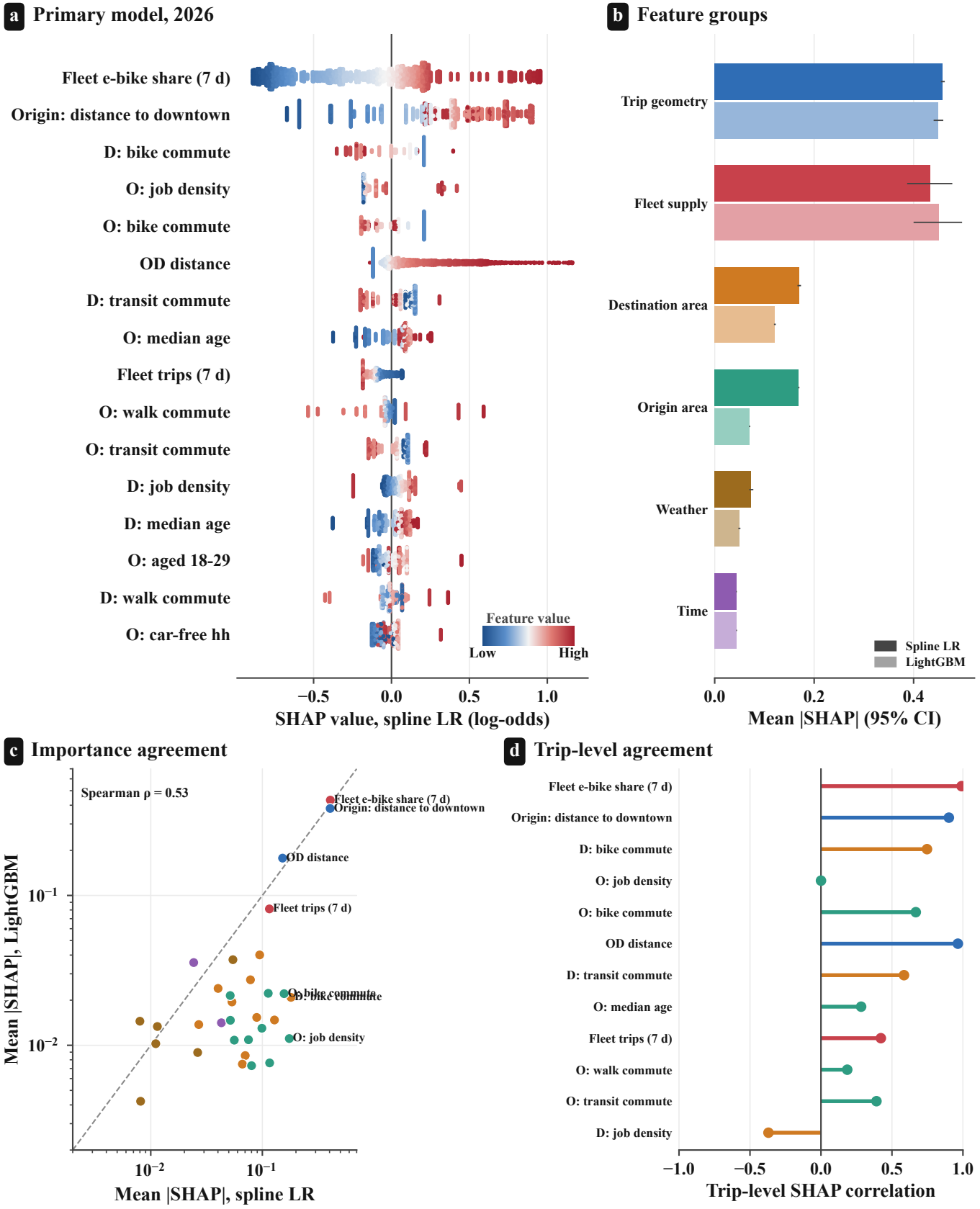

a Primary model, 2026
Fleet e-bike share (7 d)
Origin: distance to downtown
D: bike commute
O: job density
O: bike commute
OD distance
D: transit commute
O: median age
Fleet trips (7 d)
O: walk commute
O: transit commute
D: job density
D: median age
O: aged 18-29
D: walk commute
O: car-free hh
Feature value
Low
High
−0.5
0.0
0.5
1.0
SHAP value, spline LR (log-odds)
b Feature groups
Trip geometry
Fleet supply
Destination area
Origin area
Weather
Time
Spline LR
LightGBM
0.0
0.2
0.4
Mean |SHAP| (95% CI)
c Importance agreement
Spearman ρ = 0.53
Fleet e-bike share (7 d)
Origin: distance to downtown
OD distance
Fleet trips (7 d)
O: bike commute
D: bike commute
O: job density
10^{-1}
10^{-2}
Mean |SHAP|, LightGBM
Mean |SHAP|, spline LR
d Trip-level agreement
Fleet e-bike share (7 d)
Origin: distance to downtown
D: bike commute
O: job density
O: bike commute
OD distance
D: transit commute
O: median age
Fleet trips (7 d)
O: walk commute
O: transit commute
D: job density
−1.0
−0.5
0.0
0.5
1.0
Trip-level SHAP correlation